

CAMBRIDGE INTERNATIONAL AS LEVEL BIOLOGY (9700)

TOPIC 9 – GAS EXCHANGE

Respiratory Anatomy, Alveolar Gas Exchange & Effects of Tobacco Smoking

CANDIDATE

Hamna (Grade A* Scholar)

SPECIFICATION

Cambridge 9700 AS Level

TOTAL MARKS / QUESTIONS

220 Marks · 50 Questions

EXAMINATION STRUCTURE & 80/20 QUESTION TARIFF

- Section A – Extended High-Tariff Questions: 20 × 6 Marks (120 Marks total) [Multi-part synoptic evaluation]
- Section B – Data Interpretation & Structured Explanations: 20 × 4 Marks (80 Marks total)
- Section C – Core Definitions & Quantitative Calculations: 10 × 2 Marks (20 Marks total)
- Section D – 10 High-Frequency Core Repeated Questions for Exam Mastery & Knowledge Consolidation
- Division: 80% Authentic Cambridge 9700 Past Paper references + 20% Original A* Extension Problems

SYLLABUS SUBTOPICS COVERED IN THIS PACK

9.1 The gas exchange system: gross anatomy of the respiratory tract (larynx, trachea, bronchi, bronchioles, terminal and respiratory bronchioles, alveoli, capillary network); histological tissue distribution (hyaline cartilage C-rings in trachea, irregular plates in bronchi, absent in bronchioles; pseudostratified ciliated columnar epithelium, goblet cells, seromucous glands, smooth muscle, elastic fibres, squamous epithelium); recognition in light photomicrographs, electron micrographs, and plan diagrams; functional adaptations of ciliated cells and goblet cells (mucociliary escalator), cartilage (airway patency), smooth muscle (bronchomotor tone), elastic fibres (alveolar expansion and passive recoil), and squamous epithelium (ultrathin diffusion barrier)

9.2 Gas exchange at the alveolar surface: quantitative application of Fick's first law of diffusion ($\text{Rate} = A \times \Delta P / x$); physiological adaptations maximising surface area (~100 m² across 500–700 million alveoli, extensive branching capillary meshwork); maintaining steep partial pressure gradients (alveolar pO₂ = 104 mmHg, capillary pO₂ = 40 mmHg; alveolar pCO₂ = 40 mmHg, capillary pCO₂ = 45 mmHg) via continuous ventilation and continuous pulmonary capillary perfusion; ultrathin diffusion barrier (< 0.5 µm: Type I pneumocyte cytoplasm, shared fused basement membrane, endothelial cell cytoplasm); single-file RBC transit; pulmonary surfactant biophysics (Type II pneumocytes, DPPC, surface tension reduction, prevention of alveolar collapse / atelectasis and IRDS)

9.3 Tobacco smoking impacts and pathology: triad of tobacco toxins (nicotine, carbon monoxide, tar); nicotine mode of action (sympathetic stimulation, adrenaline release, arteriolar vasoconstriction, hypertension, platelet activation, atheroma and thrombosis); carbon monoxide toxicity (irreversible binding to haemoglobin with 250x higher affinity forming carboxyhaemoglobin HbCO, leftward shift of dissociation curve, tissue hypoxia, fetal growth restriction); tar actions (ciliostasis, goblet cell hyperplasia, mucus hypersecretion); pathogenesis of respiratory diseases: chronic bronchitis (chronic inflammation, productive smoker's cough, bacterial superinfections), pulmonary emphysema (macrophage/neutrophil elastase secretion, oxidative inactivation of alpha₁-antitrypsin, proteolytic destruction of alveolar septa, loss of elastic recoil, air trapping, bullae formation), COPD, and multi-step bronchogenic carcinoma (benzo[a]pyrene DNA adducts, TP53 and KRAS mutations, dysplasia, invasive carcinoma, metastasis)

EXAMINER'S DIRECTIVE FOR CANDIDATE HAMNA

Write legibly in the dotted spaces provided. Pay strict attention to Cambridge command words (Explain, Describe, Deduce, Calculate).

Where biological diagrams or graphs are provided, extract exact qualitative and quantitative coordinates to support your answers.

1 9700/22/M/J/23/Q3 – Gross Anatomy and Functional Organisation of the Human Gas Exchange System [6 Marks] Syllabus 9.1

The human gas exchange system consists of conducting airways and respiratory surfaces adapted for the efficient uptake of oxygen and removal of carbon dioxide. Fig. 9.1 illustrates the gross anatomy of the respiratory tract from the larynx to the terminal alveoli.

Fig. 9.1: Gross Anatomy of the Human Gas Exchange System

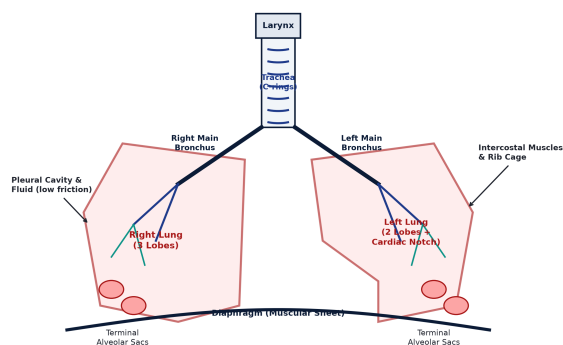


Fig. 9.1: Gross anatomy of the human gas exchange system showing conducting airways, lobar lungs, and diaphragm.

- (a) Describe the route taken by an inspired oxygen molecule from the larynx to an alveolar air space, naming each conducting airway in sequence. [2]
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- (b) Explain the physiological importance of the incomplete C-shaped cartilage rings in the trachea compared to complete circular rings. [2]
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- (c) Describe the role of the pleural fluid and pleural membranes surrounding each lung during ventilation. [2]
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2 9700/21/O/N/22/Q2 – Histological Structure of the Trachea Wall and Function of Component Tissues [6 Marks] Syllabus 9.1 & 9.2

The wall of the trachea contains several distinct tissue layers that support the airway and protect against inhaled pathogens. Fig. 9.2 is a histological plan diagram of a transverse section (TS) through the tracheal wall.

Fig. 9.2: Histological Plan Diagram of Trachea Wall (Transverse Section)

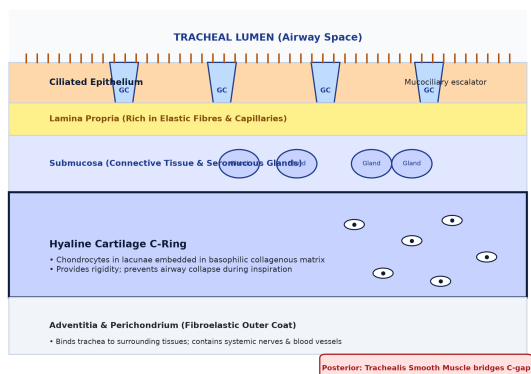


Fig. 9.2: Histological plan diagram of a transverse section through the anterior wall of the human trachea.

- (a) Identify the tissue forming the hyaline cartilage ring and describe its cellular organisation and extracellular matrix. [2]
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- (b) State the functions of the seromucous glands located in the submucosa and the elastic fibres in the lamina propria. [2]
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- (c) Describe the structure and location of the trachealis muscle and outline its function during coughing. [2]
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3 9700/22/M/J/22/Q4 – Ultrastructure of Ciliated Columnar Epithelium and the Mucociliary Escalator [6 Marks] Syllabus 9.1 & 9.2

The pseudostratified columnar epithelium lining the trachea and bronchi functions as a vital defensive barrier. Fig. 9.3 shows the fine ultrastructure of ciliated epithelial cells, goblet cells, and the overlying mucus blanket.

Fig. 9.3: Ultrastructure of Ciliated Epithelium, Goblet Cells & Mucus Layer

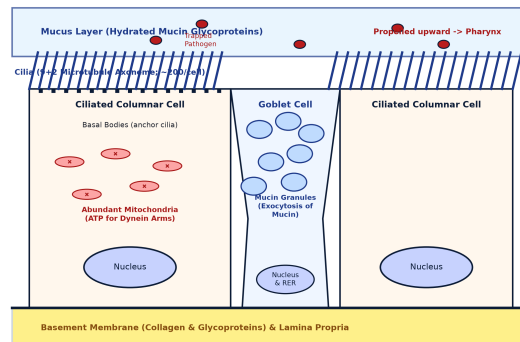


Fig. 9.3: Transmission electron micrograph schematic of ciliated epithelial cells and a goblet cell forming the mucociliary escalator.

- (a) Describe the internal microtubule arrangement of a cilium and outline how dynein motor arms produce ciliary beating. [2]
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- (b) Explain why ciliated epithelial cells contain a large number of mitochondria clustered beneath the basal bodies. [2]
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- (c) Describe the synthesis and secretion of mucin glycoproteins by goblet cells and outline the function of the mucus layer. [2]
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4 9700/23/O/N/21/Q2 – Comparative Histology of Bronchi and Bronchioles: Tissue Distribution and Function [6 Marks] Syllabus 9.1 & 9.2

As the bronchial tree branches deeper into the lung parenchyma, marked histological transitions occur. Fig. 9.4 compares transverse sections of a medium bronchus and a terminal bronchiole.

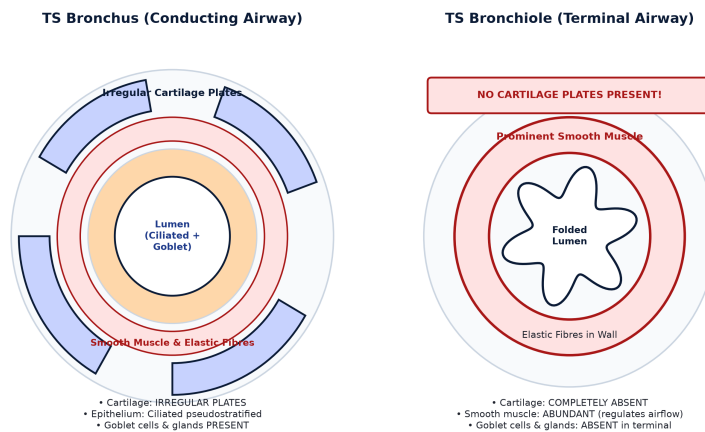


Fig. 9.4: Comparison of transverse sections (TS) through a bronchus and a terminal bronchiole.

- (a) State two major structural differences visible in Fig. 9.4 between the wall of a bronchus and the wall of a bronchiole. [2]

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- (b) Explain how the prominent ring of smooth muscle in the bronchiole wall regulates ventilation to different pulmonary regions. [2]

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- (c) Explain why the absence of cartilage in bronchioles makes them vulnerable to collapse in patients with severe emphysema. [2]

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5 9700/22/F/M/23/Q3 – Ultrastructure of the Alveolar–Capillary Unit and the Diffusion Barrier

[6 Marks] Syllabus 9.1 & 9.3

Gas exchange between alveolar air and pulmonary capillary blood occurs across an extremely thin barrier. Fig. 9.5 illustrates the ultrastructural components of the alveolar–capillary barrier and the cellular components of the alveolar septum.

Fig. 9.5: Ultrastructure of the Alveolar–Capillary Gas Exchange Barrier

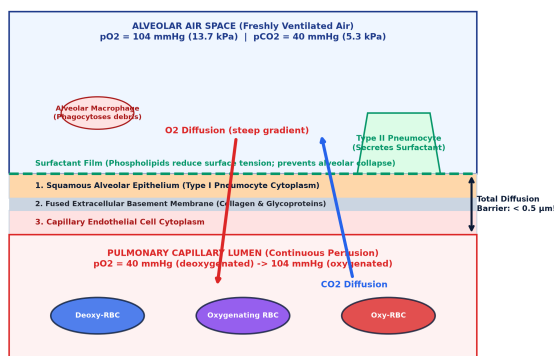


Fig. 9.5: Ultrastructure of the alveolar–capillary interface showing pneumocytes, basement membranes, and endothelial cells.

- (a) List the three specific structural layers that comprise the gas exchange barrier between alveolar air and capillary blood plasma. [2]
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- (b) Distinguish between the structure and function of Type I pneumocytes and Type II pneumocytes. [2]
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- (c) Explain the physiological consequence if Type II pneumocytes fail to secrete sufficient pulmonary surfactant. [2]
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6 9700/21/M/J/23/Q4 – Role of Alveolar Elastic Fibres in Ventilation and Pathological Elastic Breakdown [6 Marks] Syllabus 9.1 & 9.2

Alveoli are enveloped in a dense basket-like network of elastic fibres consisting of elastin proteins. Fig. 9.6 compares the physical state of these elastic fibres during inhalation and exhalation.

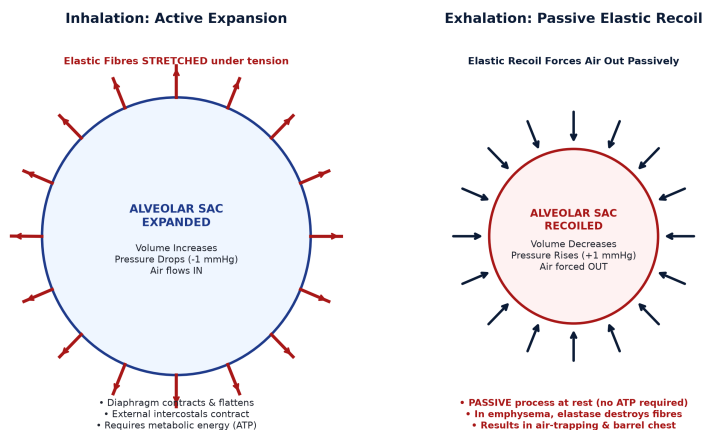


Fig. 9.6: Schematic diagram showing the stretching of alveolar elastic fibres during inspiration and passive recoil during expiration.

- (a) Explain the mechanical role of elastic fibres during inspiration and quiet expiration at rest. [2]
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- (b) State why expiration at rest is described as an energetically passive process. [2]
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- (c) Explain how the destruction of elastic fibres in a chronic smoker leads to the development of a 'barrel chest'. [2]
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7 9700/22/O/N/23/Q4 – Analysis of Spirometry Traces: Lung Volumes, Capacities, and Ventilation Rate

[6 Marks] Syllabus 9.1

A spirometer was used to record changes in lung volume of a healthy 18-year-old student at rest and during maximal respiratory efforts. Fig. 9.7 shows the resulting spirogram trace.

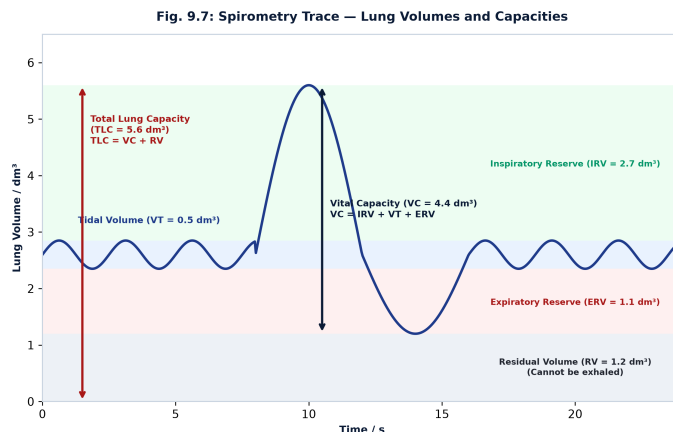


Fig. 9.7: Spirometer trace showing tidal breathing, maximal inspiration, maximal expiration, and residual volume.

- (a) Using values from Fig. 9.7, determine the Tidal Volume (V_T) and calculate the student's Vital Capacity (V_C). [2]

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- (b) Define Residual Volume (R_V) and explain why it cannot be measured directly using a simple water-filled spirometer. [2]

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- (c) The student had a resting breathing rate of 14 breaths per minute. Calculate the Pulmonary Ventilation Rate (PVR) in $\text{dm}^3 \text{ min}^{-1}$, showing your working. [2]

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8 9700/21/M/J/21/Q3 – Fick's First Law of Diffusion and Quantitative Adaptations of the Alveolar Surface

[6 Marks] Syllabus 9.3

The rate of gas exchange across pulmonary membranes is governed by physical laws of diffusion. Fig. 9.8 summarizes Fick's first law of diffusion and the specific physiological factors operating at the human alveolar membrane.

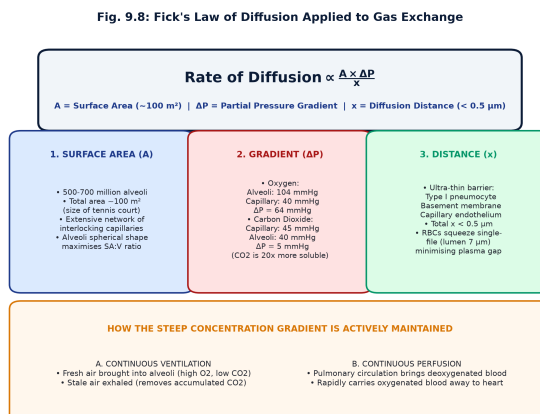


Fig. 9.8: Mathematical expression of Fick's law and physiological factors optimizing diffusion rate in lungs.

- (a) State Fick's first law of diffusion in words or as a proportional relationship, defining each variable. [2]
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- (b) Explain how the respiratory and circulatory systems cooperate to maintain a steep partial pressure gradient for oxygen. [2]
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- (c) Although the partial pressure gradient for CO₂ (5 mmHg) is much smaller than for O₂ (64 mmHg), both gases diffuse at comparable total rates. Explain why. [2]
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9 9700/22/M/J/20/Q3 – Pathogenesis and Histopathological Hallmarks of Chronic Bronchitis

[6 Marks] Syllabus 9.4 & 9.5

Tobacco smoke contains irritants that provoke progressive structural damage to the bronchial tree. Fig. 9.9 compares the histology of healthy bronchial mucosa with that from a long-term cigarette smoker suffering from chronic bronchitis.

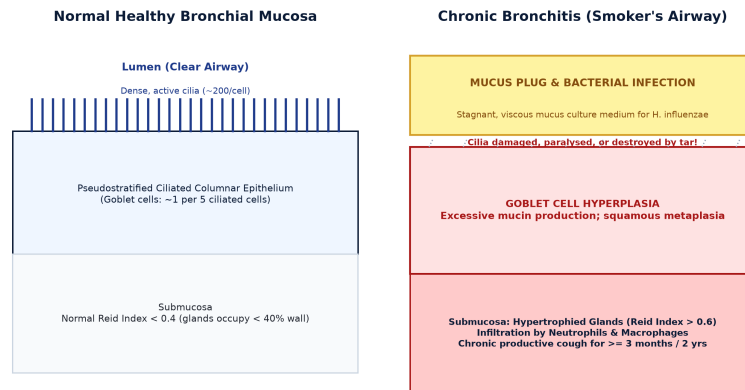


Fig. 9.9: Histopathological alterations in the bronchial epithelium and submucosa in chronic bronchitis compared to healthy tissue.

- (a) Describe the cellular changes that occur in the epithelial lining of the bronchi in chronic bronchitis. [2]
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- (b) Explain why chronic bronchitis patients suffer from a persistent, productive cough ('smoker's cough'). [2]
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- (c) Explain how chronic bronchitis increases susceptibility to recurrent secondary bacterial infections. [2]
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10 9700/22/O/N/20/Q4 – Molecular and Cellular Pathogenesis of Pulmonary Emphysema: The Protease-Antiprotease Hypothesis

Emphysema is characterized by irreversible destruction of alveolar walls. Fig. 9.10 outlines the biochemical pathway involving inflammatory cells, proteolytic enzymes, and protective antiproteases in emphysema.

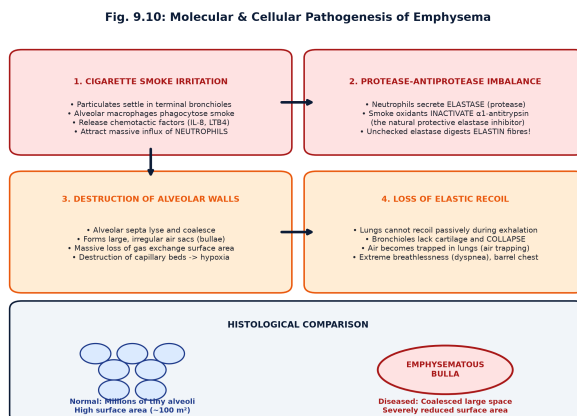


Fig. 9.10: The protease-antiprotease imbalance mechanism leading to alveolar wall destruction in pulmonary emphysema.

(a) Explain how cigarette smoke particulates trigger the recruitment and activation of neutrophils in lung tissue. [2]

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(b) Describe the normal function of alpha1-antitrypsin and explain how components of cigarette smoke impair its activity. [2]

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(c) Explain why the destruction of alveolar septa leads to severe breathlessness (dyspnea) during mild exertion. [2]

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11 9700/21/O/N/19/Q4 – Toxicological Actions of Nicotine, Carbon Monoxide, and Tar on Cardiovascular and Respiratory Systems

Cigarette smoke contains thousands of chemical compounds, three of which have distinct toxicological effects. Fig. 9.11 summarizes the modes of action of nicotine, carbon monoxide, and tar.

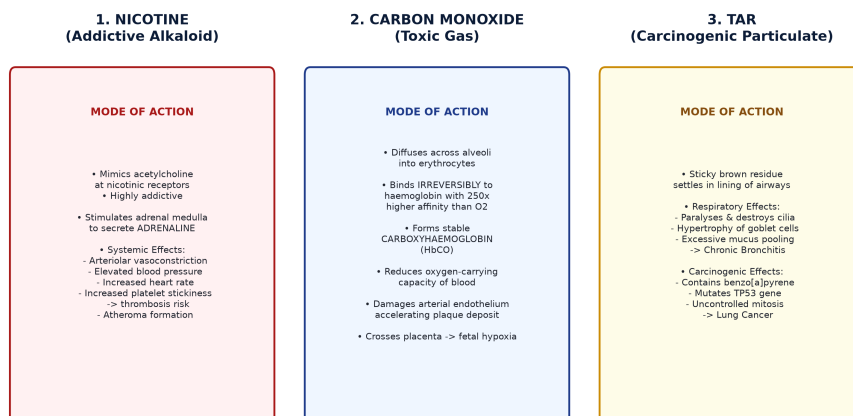


Fig. 9.11: Summary diagram comparing the chemical nature and physiological impacts of nicotine, carbon monoxide, and tar.

(a) Describe how nicotine acts on the autonomic nervous system and endocrine glands to increase cardiovascular strain. [2]

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(b) Explain the biochemical effect of carbon monoxide on haemoglobin and state how this affects fetal development in pregnant smokers. [2]

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(c) Explain how tar causes both mechanical failure of the mucociliary system and genetic mutations leading to neoplasia. [2]

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12 9700/22/M/J/19/Q3 – Multi-Step Carcinogenesis of Lung Cancer: From Metaplasia to Metastatic Carcinoma [10 marks] Syllabus 9.4 & 9.5

Lung cancer (bronchogenic carcinoma) develops through a multi-stage progression of genetic and histological changes. Fig. 9.12 illustrates the sequential stages from normal respiratory mucosa to invasive carcinoma.

Fig. 9.12: Multi-Step Pathogenesis of Bronchogenic Carcinoma (Lung Cancer)

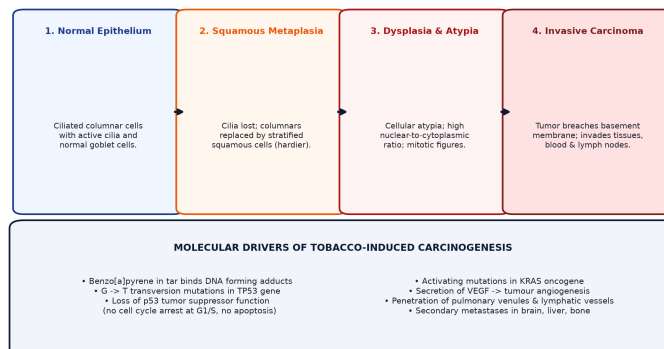


Fig. 9.12: Stages in bronchial epithelial transformation: normal mucosa → squamous metaplasia → dysplasia → invasive carcinoma.

- (a) Define the terms 'carcinogen' and 'metastasis' in the context of bronchogenic carcinoma. [2]
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- (b) Explain the roles of mutated proto-oncogenes and tumor suppressor genes in the uncontrolled proliferation of bronchial cells. [2]
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- (c) State two common clinical symptoms of lung cancer and explain the pathological basis for each symptom. [2]
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13 9700/22/M/J/18/Q3 – Thoracic Mechanics, Intrapleural Pressure, and Boyle's Law During Ventilation

[6 Marks] Syllabus 9.1

Ventilation is driven by pressure differences generated between the atmosphere and the alveolar air spaces according to Boyle's law. Fig. 9.13 outlines the neuromuscular actions and physical changes during inspiration and expiration.

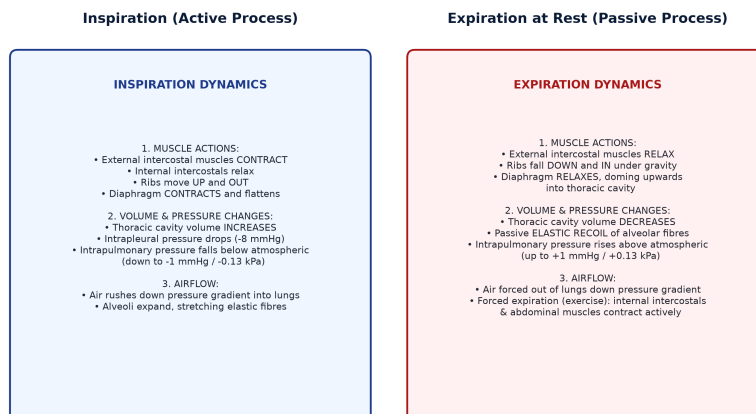


Fig. 9.13: Neuromuscular mechanisms and thoracic volume/pressure relationships during inhalation and exhalation.

(a) Describe the coordinated actions of the diaphragm and external intercostal muscles during quiet inspiration. [2]

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(b) Explain how the muscle contractions described in (a) lead to air flowing into the alveoli, with reference to Boyle's law. [2]

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(c) Describe how forced expiration (e.g. during vigorous exercise) differs from quiet expiration at rest. [2]

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14 9700/21/O/N/18/Q3 – Structural Distribution of Tissues Along the Respiratory Tract and Functional Adaptations [10 marks] Syllabus 9.1 & 9.2

The histological composition of the human respiratory tract undergoes a progressive transition from the upper conducting airways to the terminal gas exchange units. Fig. 9.14 summarizes the distribution of five key tissues across five anatomical regions.

Fig. 9.14: Distribution of Tissues Across the Human Gas Exchange System

Region	Cartilage	Ciliated Epithelium	Goblet Cells	Smooth Muscle	Elastic Fibres
Trachea	C-shaped Rings	Abundant (Pseudostratified)	Abundant	Trachealis (Dorsal)	Abundant in Lamina Propria
Bronchus	Irregular Plates	Abundant (Columnar)	Present	Interlacing Bands	Abundant
Terminal Bronchiole	ABSENT	Ciliated Cuboidal	Very Few / Absent	Complete Circular Ring	Abundant
Respiratory Bronchiole	ABSENT	Cuboidal (Few Cilia)	ABSENT	Scattered Bundles	Abundant
Alveolus	ABSENT	ABSENT (Squamous Type I)	ABSENT	ABSENT	Dense Network (Recoil)

CRITICAL CAMBRIDGE EXAMINER MANDATE: Cartilage is NEVER present in bronchioles!
Elastic fibres are present throughout the entire respiratory tract from trachea to alveoli.

Fig. 9.14: Tissue distribution matrix showing cartilage, ciliated epithelium, goblet cells, smooth muscle, and elastic fibres.

- (a) With reference to Fig. 9.14, state which tissue is present in every region from the trachea to the alveoli, and explain why this tissue is essential throughout. [2]
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- (b) Describe the transition in epithelial cell morphology from the trachea to the alveoli. [2]
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- (c) Explain why goblet cells and cartilage must both be absent from the alveolar walls. [2]
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15 9700/22/O/N/22/Q3 – Experimental Investigation of Cigarette Smoke Condensates Using a Model Lung Apparatus [10 marks] Syllabus 9.4

A smoking machine was set up in which the smoke of five burning cigarettes was drawn through glass tubing packed with white glass wool / cotton wool, followed by universal indicator solution and limewater.

- (a) Describe and explain the appearance of the glass wool after drawing the smoke of five cigarettes through the apparatus. [2]
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- (b) Predict and explain the colour change observed in the universal indicator solution, naming the chemical species responsible. [2]
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- (c) Explain why the limewater becomes cloudy, and evaluate one limitation of this apparatus as a physiological model for human inhalation. [2]
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16 9700/21/M/J/22/Q2 – Kinetics of Carbon Monoxide Binding to Haemoglobin and Tissue Hypoxia

[6 Marks] Syllabus 9.4 & 8.2

Carbon monoxide is a colourless, odourless toxic gas produced by incomplete combustion of organic matter, including tobacco. Blood samples from non-smokers and heavy smokers were analysed for carboxyhaemoglobin (HbCO) percentage.

- (a) Explain why even low atmospheric concentrations of carbon monoxide (e.g. 0.1%) result in dangerous levels of HbCO in blood. [2]

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- (b) Explain how the formation of HbCO alters the affinity of the remaining unaffected haem groups for oxygen (allosteric effect). [2]

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- (c) Suggest why heavy smokers experience elevated resting hematocrit (red blood cell count) and higher blood viscosity. [2]

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17 9700/23/M/J/20/Q3 – Differential Diagnosis and Pathophysiology of Chronic Obstructive Pulmonary Disease (COPD)

[6 Marks] Syllabus 9.5

Chronic Obstructive Pulmonary Disease (COPD) is an umbrella term encompassing chronic bronchitis and emphysema, which often coexist in long-term cigarette smokers.

- (a) Compare chronic bronchitis and emphysema with respect to the primary anatomical site of disease within the lung. [2]

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- (b) Explain why patients with pure emphysema are often termed 'pink puffers' whereas advanced chronic bronchitis patients are termed 'blue bloaters'. [2]

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- (c) Explain how long-standing COPD can lead to pulmonary hypertension and subsequent right ventricular heart failure (cor pulmonale). [2]

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18 9700/21/M/J/17/Q3 – Modelling Gas Diffusion Rates and the Surface Area-to-Volume Ratio Dilemma

[6 Marks] Syllabus 9.3 & 4.2

A student investigated the relationship between organism size and diffusion by immersing agar cubes containing phenolphthalein indicator and dilute sodium hydroxide into dilute hydrochloric acid.

- (a) State the formula for calculating surface area-to-volume ratio (SA:V) of a cube with side length l , and explain what happens to SA:V as size increases. [2]

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- (b) Explain why the absolute distance diffused by the acid is identical in all cubes after 10 minutes, but the percentage volume decolourised is lowest in the largest cube. [2]

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- (c) With reference to your answer in (a) and (b), explain why large multicellular mammals cannot rely on external body surface diffusion and require specialised lungs. [2]

19 9700/22/F/M/21/Q2 – Epidemiological Evidence and the Causal Relationship Between Smoking and Lung Cancer Syllabus 9.4 & 9.5

In 1950, Richard Doll and Austin Bradford Hill published a landmark study investigating the smoking habits of patients admitted to London hospitals with lung carcinoma compared to matched controls.

- (a) Explain why a correlation between smoking prevalence and lung cancer incidence does not automatically prove causation. [2]
- (b) Describe two lines of experimental or biochemical evidence that established cigarette smoke as the causal agent of bronchogenic carcinoma. [2]
- (c) Epidemiological data shows a 20- to 25-year lag period between increases in national cigarette consumption and rises in lung cancer mortality. Explain this lag phase. [2]

20 9700/23/O/N/20/Q3 – Biophysics of Pulmonary Surfactant, the Law of Laplace, and Infant Respiratory Distress Syndrome Syllabus 9.1 & 9.3

Pulmonary surfactant is an amphipathic complex of 90% lipids (predominantly DPPC) and 10% proteins (SP-A, SP-B, SP-C, SP-D) produced by alveolar Type II pneumocytes starting around gestational week 28–32.

- (a) Describe the molecular orientation of dipalmitoylphosphatidylcholine (DPPC) molecules at the alveolar air-liquid interface. [2]
- (b) According to the Law of Laplace ($P = 2T / r$), explain why smaller alveoli would tend to collapse into larger alveoli in the absence of surfactant. [2]
- (c) Premature neonates born before week 28 often develop Infant Respiratory Distress Syndrome (IRDS). Outline two clinical interventions used to manage this condition. [2]

21 9700/22/M/J/23/Q5 – Distribution and Morphology of Cartilage in Trachea, Bronchi, and Bronchioles [4 Marks] Syllabus 9.1 & 9.2

Cartilage provides structural support to prevent airway collapse during negative intrathoracic pressures.

- (a) State the morphological form of cartilage found in the trachea compared to that found in the bronchi. [2]

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- (b) Explain why cartilage is absent from the walls of bronchioles. [2]

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22 9700/21/O/N/21/Q4 – Comparative Cytology and Physiology: Cilia vs Microvilli [4 Marks] Syllabus 9.1 & 1.2

Cilia and microvilli are both microscopic cell surface projections found on epithelial cells but have completely different structures and functions.

- (a) Contrast the cytoskeletal core of a cilium with that of a microvillus. [2]

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- (b) Explain the physiological difference between the roles of cilia in the trachea and microvilli in the ileum. [2]

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23 9700/22/F/M/22/Q3 – Dual Regulation of Airway Calibre: Smooth Muscle vs Elastic Fibres [4 Marks] Syllabus 9.2

The walls of conducting airways contain both smooth muscle and elastic fibres which play complementary roles.

- (a) Explain how smooth muscle alters the diameter of bronchioles during an asthma attack. [2]

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- (b) Explain why elastic fibres are described as passive elements, contrasting their action with smooth muscle. [2]

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24 9700/21/M/J/20/Q4 – Secretory Function of Goblet Cells and Submucosal Glands [4 Marks] Syllabus 9.2

Mucus production is a primary non-specific innate immune mechanism protecting the gas exchange surfaces.

- (a) Describe the chemical nature of mucin and explain how it forms mucus upon secretion. [2]

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- (b) Explain why excessive mucus secretion can impair gas exchange even if alveoli are undamaged. [2]

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25 9700/22/M/J/21/Q4 – Structural Adaptations of Alveoli for Rapid Diffusion

[4 Marks] Syllabus 9.3

The human gas exchange system is adapted to maximise the diffusion of oxygen into blood.

- (a) Describe two structural adaptations of alveoli that provide a high surface area for diffusion. [2]

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- (b) Explain how the capillaries surrounding alveoli contribute to a short diffusion distance. [2]

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26 9700/21/O/N/20/Q3 – Quantitative Comparison of Inspired and Expired Air Composition

[4 Marks] Syllabus 9.3

During gas exchange in the alveoli, the chemical composition of air is substantially altered.

- (a) State the approximate percentage of oxygen and carbon dioxide in inspired air and expired air. [2]

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- (b) Explain why expired air still contains approximately 16% oxygen rather than 0%. [2]

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27 9700/22/F/M/20/Q4 – Pharmacological Effects of Nicotine on Cardiovascular Hemodynamics

[4 Marks] Syllabus 9.4

Nicotine is an alkaloid absorbed within seconds through the alveolar capillaries into pulmonary blood.

- (a) Explain why nicotine absorption causes an immediate increase in heart rate and arterial blood pressure. [2]

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- (b) Explain how nicotine promotes atheroma formation and arterial thrombosis. [2]

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28 9700/22/O/N/19/Q3 – Carbon Monoxide Toxicity and the Shift in Oxygen Dissociation

[4 Marks] Syllabus 9.4

Carbon monoxide gas binds competitively to the same coordination sites on haemoglobin as oxygen.

- (a) Name the compound formed when carbon monoxide binds to haemoglobin and state why this binding is persistent. [2]

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- (b) Explain why a person with 20% carboxyhaemoglobin experiences severe fatigue during exercise. [2]

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29 9700/21/M/J/19/Q4 – Action of Tobacco Tar on Ciliated Epithelial Function

[4 Marks] Syllabus 9.4

Tobacco tar is a complex mixture of particulates and chemical carcinogens that settles in conducting airways.

- (a) Explain what is meant by 'ciliostasis' and describe how tar induces this condition. [2]

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- (b) Describe the consequence of ciliostasis for the respiratory defence mechanisms. [2]

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30 9700/22/M/J/18/Q4 – Mucus Hypersecretion and Goblet Cell Alterations in Chronic Bronchitis

[4 Marks] Syllabus 9.5

In chronic bronchitis, prolonged exposure to tobacco smoke leads to hypertrophy of secretory structures.

- (a) Distinguish between goblet cell 'hypertrophy' and 'hyperplasia'. [2]

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- (b) Explain how chronic bronchitis causes narrowing of airway lumina. [2]

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31 9700/23/O/N/18/Q2 – Structural Alterations of Alveoli in Pulmonary Emphysema

[4 Marks] Syllabus 9.5

Emphysema is characterized by anatomical changes in the gas-exchanging parenchyma of the lung.

- (a) Describe how the microscopic appearance of alveolar tissue in an emphysema patient differs from healthy lung tissue. [2]

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- (b) Explain how this change in appearance reduces the rate of oxygen uptake into blood. [2]

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32 9700/21/M/J/18/Q4 – The Enzymatic Role of Elastase and its Physiological Regulation

[4 Marks] Syllabus 9.5 & 3.2

Under healthy conditions, a balance exists between destructive proteases and protective antiproteases in lung tissue.

- (a) Identify the leukocyte that releases elastase into lung tissues and state its physiological purpose during infection. [2]

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- (b) Explain why individuals with genetic alpha₁-antitrypsin deficiency develop severe emphysema at an early age even if they do not smoke. [2]

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33 9700/22/F/M/19/Q4 – Chemical Carcinogens in Tobacco Smoke and Mutagenic Mechanisms

[4 Marks] Syllabus 9.4 & 5.1

Chemical analysis of tobacco tar identifies polycyclic aromatic hydrocarbons such as benzo[a]pyrene.

- (a) Describe how benzo[a]pyrene interacts with DNA in bronchial epithelial cells. [2]

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- (b) Explain how a mutation in the TP53 gene contributes to the formation of a malignant lung tumor. [2]

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34 9700/21/O/N/17/Q3 – Neuromuscular Coordination During Inhalation

[4 Marks] Syllabus 9.1

Inhalation requires active contraction of skeletal muscle groups innervated by the phrenic and intercostal nerves.

- (a) Describe the change in shape of the diaphragm upon contraction and state how this affects thoracic cavity volume. [2]

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- (b) Explain why air enters the lungs passively following thoracic expansion. [2]

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35 9700/22/M/J/17/Q4 – Comparative Mechanics: Resting Expiration vs Active Forced Expiration

[4 Marks] Syllabus 9.1

The muscular mechanism of expiration changes dramatically between resting conditions and strenuous physical activity.

- (a) State which muscles are active during quiet expiration at rest. [2]

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- (b) Name two muscle groups that contract during forced expiration and describe their mechanical effect. [2]

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36 9700/23/M/J/17/Q3 – Biochemical Composition and Biomechanical Role of Pulmonary Surfactant

[4 Marks] Syllabus 9.3

Alveoli are lined with a thin film of water which generates inward surface tension forces.

- (a) Explain how water molecules generate surface tension at the alveolar air-liquid interface. [2]

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- (b) Describe how surfactant molecules reduce this surface tension and state the physiological advantage. [2]

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37 9700/21/O/N/16/Q3 – Pulmonary Functional Volumes: Tidal Volume and Vital Capacity

[4 Marks] Syllabus 9.1

Spirometric measurements provide objective indicators of pulmonary function.

- (a) Define Tidal Volume (VT) and state its typical value in a healthy young adult at rest. [2]

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- (b) Define Vital Capacity (VC) and write a formula expressing VC in terms of its component lung volumes. [2]

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38 9700/22/M/J/16/Q4 – Microcirculatory Hemodynamics in Alveolar Capillaries

[4 Marks] Syllabus 9.3 & 8.1

Erythrocytes traverse pulmonary capillaries in less than one second under resting conditions.

- (a) State the typical transit time of an erythrocyte through a pulmonary capillary and explain why full oxygen saturation occurs within 0.25 s. [2]

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- (b) Explain the physiological benefit of erythrocytes having to deform and squeeze single-file through capillaries. [2]

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39 9700/21/M/J/16/Q3 – Health Consequences of Passive Smoking on Non-Smokers

[4 Marks] Syllabus 9.4

Passive smoking refers to the involuntary inhalation of secondhand tobacco smoke by non-smokers.

- (a) Distinguish between 'sidestream smoke' and 'mainstream smoke', and explain why sidestream smoke can be more hazardous. [2]

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- (b) Describe two documented health impacts of passive smoking in young children living with smokers. [2]

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40 9700/22/F/M/16/Q3 – Cellular Defense of the Alveolar Space: The Alveolar Macrophage

[4 Marks] Syllabus 9.1 & 11.1

Because alveoli lack cilia and goblet cells, particulate matter that reaches the terminal air spaces must be cleared by specialized phagocytes.

- (a) Describe the origin and location of alveolar macrophages ('dust cells'). [2]

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- (b) Explain how alveolar macrophages process inhaled carbon particulates and what happens when they become overwhelmed. [2]

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41 9700/22/M/J/23/Q1(a) – Rigorous Scientific Definition: Gas Exchange

[2 Marks] Syllabus 9.1

- (a) Define the biological term 'gas exchange' as it applies to mammals.

[2]

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42 9700/21/O/N/22/Q1(a) – Essential Distinction: Ventilation vs Cellular Respiration

[2 Marks] Syllabus 9.1

- (a) Distinguish clearly between the terms 'ventilation' (breathing) and 'cellular respiration'.

[2]

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43 9700/22/M/J/22/Q1(b) – Functional Significance of Incomplete Cartilage Rings

[2 Marks] Syllabus 9.1

- (a) State the function of the C-shaped cartilage rings in the wall of the mammalian trachea.

[2]

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44 9700/23/M/J/21/Q1(c) – Mechanism of Passive Alveolar Recoil

[2 Marks] Syllabus 9.2

- (a) Explain the function of elastic fibres in the alveolar walls during quiet expiration.

[2]

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45 9700/21/O/N/21/Q1(a) – Definition and Normal Value: Tidal Volume

[2 Marks] Syllabus 9.1

- (a) Define the term 'tidal volume' and state its standard resting value in an adult human.

[2]

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46 9700/22/F/M/21/Q1(b) – Definition: Vital Capacity

[2 Marks] Syllabus 9.1

- (a) Define the term 'vital capacity' of the lungs.

[2]

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47 9700/21/M/J/20/Q1(a) – Diagnostic Definition: Chronic Bronchitis

[2 Marks] Syllabus 9.5

- (a) State the clinical diagnostic definition of chronic bronchitis in terms of symptoms and duration.

[2]

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48 9700/22/O/N/19/Q1(a) – Pathological Definition: Pulmonary Emphysema

[2 Marks] Syllabus 9.5

- (a) State the primary histopathological defect that defines pulmonary emphysema.

[2]

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49 9700/21/M/J/19/Q1(b) – Chemical Nature: Carboxyhaemoglobin

[2 Marks] Syllabus 9.4

- (a) Define 'carboxyhaemoglobin' and state why it causes cellular hypoxia.

[2]

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50 9700/22/M/J/18/Q1(a) – Rigorous Definition: Carcinogen

[2 Marks] Syllabus 9.4

- (a) Define the term 'carcinogen' and name one specific chemical carcinogen present in tobacco tar.

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SECTION D: 10 HIGH-FREQUENCY EXAMINER FAQs & CRITICAL REVISION PITFALLS

Essential Cambridge International AS Level Biology (9700) Mark-Scheme Traps, Core Misconceptions, and Model Answers

HISTOLOGY • CYTOSKELETAL ORGANELLES

FAQ 1: Why must candidates strictly distinguish between cilia and microvilli in histology?

Cambridge Examiner Warning: CRITICAL EXAMINER TRAP: Candidates constantly confuse cilia with microvilli! Writing that 'cilia increase surface area for gas absorption' or that 'microvilli sweep mucus' loses all marks. Cilia have a 9+2 microtubule axoneme with dynein motor arms and beat actively; microvilli are non-motile actin core folds.

- **Structure of Cilia:** Motile hair-like extensions of the cell surface membrane containing an internal axoneme consisting of nine peripheral doublet microtubules surrounding two central singlet microtubules ('9+2' arrangement) anchored to a basal body.
- **Mechanism of Motility:** Dynein ATPase arms hydrolyse ATP to cause adjacent microtubule doublets to slide past each other, generating coordinated, wave-like metachronal beating.
- **Function of Cilia:** In the trachea, bronchi, and larger bronchioles, cilia beat in synchrony (~10–15 Hz) to propel the overlying sticky mucus blanket upwards towards the pharynx and esophagus (mucociliary escalator), clearing trapped particulates and microorganisms.
- **Contrast with Microvilli:** Microvilli are non-motile, cylindrical finger-like plasma membrane folds supported internally by parallel actin microfilament bundles (no microtubules, no basal bodies). Their sole function is to vastly expand apical surface area for membrane transport proteins (found in small intestine enterocytes and kidney proximal convoluted tubules, NOT in the tracheal respiratory lining).

AIRWAY ANATOMY • TISSUE DISTRIBUTION

FAQ 2: What is the exact distribution of cartilage, smooth muscle, and elastic fibres from trachea to alveoli?

Cambridge Examiner Warning: CRITICAL EXAMINER TRAP: Stating that cartilage is present in bronchioles or alveoli! Cartilage is C-shaped in the trachea, irregular plates in bronchi, and completely ABSENT in bronchioles. Bronchioles have a prominent circular layer of smooth muscle and abundant elastic fibres.

- **Trachea:** Possesses 16–20 C-shaped (incomplete) hyaline cartilage rings joined posteriorly by the smooth trachealis muscle. Lamina propria contains elastic fibres; submucosa contains seromucous glands. Lined by pseudostratified ciliated columnar epithelium with abundant goblet cells.
- **Bronchi:** Possesses irregular curved plates / islands of hyaline cartilage embedded in the wall. Smooth muscle forms criss-crossing spiral bands between cartilage and mucosa. Ciliated columnar epithelium and goblet cells remain present.
- **Bronchioles (< 1 mm diameter):** Completely lack cartilage plates! Wall is dominated by a complete, thick circular coat of smooth muscle and abundant surrounding elastic fibres. Lined initially by ciliated cuboidal epithelium; goblet cells and seromucous glands are entirely absent from terminal and respiratory bronchioles.
- **Alveoli:** Consist purely of an extremely thin simple squamous epithelium (Type I pneumocytes) and surfactant-secreting Type II pneumocytes. Completely lack cartilage, smooth muscle, and cilia. Enveloped externally by an extensive network of elastic fibres and pulmonary capillaries.

PULMONARY MECHANICS • ELASTIC RECOIL

FAQ 3: Why do examiners penalise stating that elastic fibres 'contract' during exhalation?

Cambridge Examiner Warning: CRITICAL EXAMINER TRAP: Saying elastic fibres 'contract' to push air out! Proteins like elastin have no ATPase activity and never contract actively. Elastic fibres stretch under tension during inhalation and RECOIL passively during exhalation. Only muscle contracts actively.

- **Passive Nature of Elastic Fibres:** Elastic fibres are composed of the extracellular fibrous protein elastin cross-linked into a random-coil network. When thoracic volume expands during inspiration, external mechanical work stretches these fibres, storing potential energy.
- **Passive Elastic Recoil:** During quiet exhalation at rest, inspiratory muscles (diaphragm and external intercostals) relax. The stretched elastic fibres spontaneously recoil back to their original resting length, squeezing the alveolar air space and forcing air outwards down a positive pressure gradient without any ATP consumption.
- **Active Action of Smooth Muscle:** In contrast, airway smooth muscle cells contain actin and myosin myofilaments that actively hydrolyse ATP to contract, causing bronchoconstriction (e.g. in response to cold air, allergens, or parasympathetic acetylcholine), and actively relax under sympathetic stimulation (noradrenaline / adrenaline) to cause bronchodilation.

FICK'S LAW • DIFFUSION BARRIER

FAQ 4: What are the exact components of the alveolar-capillary diffusion barrier?

Cambridge Examiner Warning: CRITICAL EXAMINER TRAP: Writing that the diffusion barrier is 'one cell thick' or just 'alveolar cell and capillary'. Cambridge mark schemes strictly award marks only for identifying the THREE distinct anatomical layers: squamous alveolar cell (Type I pneumocyte), fused extracellular basement membrane, and capillary endothelial cell.

- **Layer 1 (Alveolar Epithelium):** The extremely thin cytoplasm of a Type I pneumocyte (simple squamous alveolar cell), approximately 0.1–0.2 µm thick.
- **Layer 2 (Basement Membrane):** The shared, fused extracellular basement membrane composed of type IV collagen and laminin glycoproteins produced by the epithelial and endothelial cells, approximately 0.1 µm thick.
- **Layer 3 (Capillary Endothelium):** The cytoplasm of the capillary endothelial cell, approximately 0.1–0.2 µm thick.
- **Total Diffusion Distance:** The combined barrier thickness is less than 0.5 µm (typically 0.2–0.4 µm), providing an exceptionally short diffusion pathway (x) that maximizes diffusion rate according to Fick's first law.

VENTILATION-PERFUSION • CONCENTRATION GRADIENTS

FAQ 5: How is a steep concentration gradient for oxygen maintained across the alveolar membrane?

Cambridge Examiner Warning: CRITICAL EXAMINER TRAP: Saying 'there is a high gradient' without explaining the physiological mechanisms that create and sustain it. You must explicitly link ventilation (which refreshes alveolar gas) with continuous capillary perfusion (which carries oxygen away).

- Role of Continuous Ventilation: Regular tidal ventilation continuously replaces stale, oxygen-depleted alveolar air with fresh atmospheric air (containing 20.9% O₂), keeping the alveolar partial pressure of oxygen consistently high at approximately 104 mmHg (13.7 kPa).
- Role of Continuous Perfusion: The pulmonary circulation delivers deoxygenated venous blood from the right ventricle with a low partial pressure of oxygen (pO₂ ~40 mmHg / 5.3 kPa). As blood flows through the capillary, haemoglobin rapidly binds oxygen; the continuous movement of blood carries newly oxygenated erythrocytes away towards the left atrium, preventing local equilibrium and keeping capillary pO₂ low at the arterial end.
- Steep Partial Pressure Gradient: This maintains a substantial partial pressure difference ($\Delta P = 104 - 40 = 64 \text{ mmHg} / 8.4 \text{ kPa}$) driving rapid, continuous passive diffusion of O₂ into blood.

PATHOLOGY • TOBACCO SMOKING

FAQ 6: What is the precise cellular and molecular pathogenesis of chronic bronchitis?

Cambridge Examiner Warning: CRITICAL EXAMINER TRAP: Giving a superficial answer like 'mucus builds up and cough happens'. Candidates must explain the cellular changes: irritants in tar stimulate goblet cell hyperplasia and hypertrophy of submucosal glands; tar paralyzes cilia (ciliostasis); mucus pools and causes persistent cough; stagnant mucus becomes a culture medium for bacteria.

- Irritation & Secretory Hyperplasia: Chemical irritants in tobacco smoke stimulate basal stem cells in the bronchial epithelium to divide, causing hyperplasia (increased number) and hypertrophy (enlarged size) of mucus-secreting goblet cells and seromucous glands (Reid index > 0.5).
- Ciliostasis & Destruction: Tobacco tar directly impairs dynein ATPase arms and destroys cilia (ciliostasis), disabling the mucociliary escalator so mucus cannot be propelled upwards.
- Mucus Pooling & Obstruction: Copious, highly viscous mucus pools in bronchi and bronchioles, obstructing airways and triggering persistent, productive 'smoker's cough' to mechanically clear the airways.
- Secondary Bacterial Infections: Stagnant mucus serves as a nutrient-rich culture medium for pathogens such as *Streptococcus pneumoniae* and *Haemophilus influenzae*. Recurrent infections provoke chronic inflammation, attracting neutrophils and macrophages, causing scar tissue formation and narrowing of airways.

EMPHYSEMA • ELASTASE VS ALPHA-1-ANTITRYPSIN

FAQ 7: How does the protease-antiprotease imbalance explain the destruction of alveoli in emphysema?

Cambridge Examiner Warning: CRITICAL EXAMINER TRAP: Claiming that cigarette smoke 'tears' or 'dissolves' alveoli directly! The destruction is ENZYMATIC: smoke irritants attract neutrophils and activate alveolar macrophages to secrete the protease elastase. Smoke oxidants inactivate the natural protective inhibitor alpha1-antitrypsin. Unopposed elastase digests elastin fibres in alveolar septa.

- Recruitment of Phagocytes: Smoke particulates irritate terminal bronchioles and alveoli, where they are engulfed by alveolar macrophages. Macrophages release chemotactic factors (IL-8, leukotrienes) that summon large numbers of circulating neutrophils into the lung interstitium.
- Release of Elastase: Neutrophils secrete the proteolytic enzyme neutrophil elastase to digest debris. Under healthy conditions, elastase is strictly neutralized by alpha1-antitrypsin (an antiprotease synthesised in the liver).
- Inactivation of alpha1-Antitrypsin: Free radicals and oxidants in cigarette smoke oxidize active-site methionine residues on alpha1-antitrypsin, abolishing its inhibitory ability and creating an overwhelming protease-antiprotease imbalance.
- Destruction of Alveolar Septa: Unchecked elastase enzymatic digestion degrades elastin fibres in alveolar walls. Alveolar septa lyse and breakdown, coalescing millions of microscopic alveoli into large, irregular air spaces (bullae) with a severely reduced surface area for gas exchange and complete loss of elastic recoil.

TOXICOLOGY • CARDIOVASCULAR IMPACT

FAQ 8: What are the distinct physiological effects of nicotine versus carbon monoxide?

Cambridge Examiner Warning: CRITICAL EXAMINER TRAP: Confusing nicotine and carbon monoxide! Nicotine affects the nervous system and adrenal glands (causing vasoconstriction, high heart rate, high blood pressure, and platelet stickiness). Carbon monoxide binds haemoglobin to form carboxyhaemoglobin (HbCO), reducing oxygen transport and causing fetal hypoxia. Nicotine does NOT bind haemoglobin!

- Nicotine Mode of Action: An addictive alkaloid that mimics acetylcholine at nicotinic acetylcholine receptors in the central and autonomic nervous systems. Stimulates sympathetic ganglia and triggers adrenaline release from the adrenal medulla. Induces peripheral arteriolar vasoconstriction, elevating systemic vascular resistance and blood pressure; increases heart rate and cardiac workload; and increases platelet stickiness, markedly raising the risk of thrombosis and coronary heart disease.
- Carbon Monoxide (CO) Mode of Action: A toxic gas that diffuses across alveolar membranes into erythrocytes and binds competitively to iron(II) in haem groups with ~250 times higher affinity than oxygen. Forms stable carboxyhaemoglobin (HbCO), which reduces the oxygen-carrying capacity of blood. Furthermore, CO binding causes an allosteric leftward shift in the remaining oxyhaemoglobin subunits, preventing oxygen unloading at respiring tissues and causing chronic cellular hypoxia.

CARCINOGENESIS • ONCOGENES & TUMOR SUPPRESSORS

FAQ 9: How do carcinogens in tobacco tar lead to bronchogenic carcinoma (lung cancer)?

Cambridge Examiner Warning: CRITICAL EXAMINER TRAP: Writing that 'tar is cancer'. Tar contains chemical carcinogens (e.g. polycyclic aromatic hydrocarbons like benzo[a]pyrene) that mutate specific genes regulating the cell cycle (proto-oncogenes converting to oncogenes; mutating TP53 tumor suppressor gene). Uncontrolled mitosis results in a malignant tumor that invades tissue and metastasizes.

- Chemical Carcinogens: Tobacco tar contains over 60 established chemical carcinogens, including polycyclic aromatic hydrocarbons (such as benzo[a]pyrene) and tobacco-specific nitrosamines (NNK).
- DNA Adduct Formation & Mutation: Metabolic activation converts benzo[a]pyrene into reactive epoxides that covalently bind to guanine bases in DNA, forming bulky DNA adducts that induce transversion mutations (GT) during DNA replication.
- Oncogenes & Tumor Suppressor Genes: Critical mutations occur in the TP53 tumor suppressor gene (loss of p53 protein means damaged cells cannot undergo G1 cell cycle arrest or apoptosis) and the KRAS proto-oncogene (activating it into a continuously signaling oncogene).
- Tumor Progression & Metastasis: Bronchial cells escape normal cell cycle checkpoints and divide uncontrollably by mitosis, forming a localized tumor (carcinoma in situ). The malignant cells secrete angiogenic factors (VEGF), breach the basement membrane, invade bronchial walls, and enter pulmonary venules and lymphatic vessels to establish distant metastases in brain, bones, and liver.

SPIROMETRY • VENTILATION CALCULATIONS

FAQ 10: How do you calculate Pulmonary Ventilation Rate (PVR) and distinguish it from Alveolar Ventilation Rate (AVR)?

Cambridge Examiner Warning: CRITICAL EXAMINER TRAP: Forgetting that a portion of each tidal breath remains in the anatomical dead space (trachea and bronchi, ~150 cm³) and never reaches the alveoli! $PVR = \text{Tidal Volume} \times \text{Breathing Frequency}$. But $\text{Alveolar Ventilation Rate (AVR)} = (\text{Tidal Volume} - \text{Dead Space}) \times \text{Breathing Frequency}$.

- Pulmonary Ventilation Rate (PVR): Total volume of air moved into and out of the respiratory tract per minute. Formula: $PVR = \text{Tidal Volume (dm}^3) \times \text{Breathing Frequency (breaths min}^{-1})$. Example: $0.5 \text{ dm}^3 \times 12 \text{ breaths min}^{-1} = 6.0 \text{ dm}^3 \text{ min}^{-1}$.
- Anatomical Dead Space (VD): Volume of conducting airways (pharynx, trachea, bronchi, bronchioles) where no gas exchange occurs, typically ~0.15 dm³ (150 cm³) in an adult.
- Alveolar Ventilation Rate (AVR): The actual volume of fresh atmospheric air that enters the gas-exchanging alveoli per minute. Formula: $AVR = (\text{Tidal Volume} - \text{Anatomical Dead Space}) \times \text{Breathing Frequency}$. Example: $(0.50 - 0.15 \text{ dm}^3) \times 12 \text{ min}^{-1} = 0.35 \text{ dm}^3 \times 12 \text{ min}^{-1} = 4.2 \text{ dm}^3 \text{ min}^{-1}$.
- Clinical Significance: Rapid, shallow breathing (tachypnea) severely compromises AVR because dead space volume represents a larger fraction of each breath, resulting in inadequate alveolar oxygenation despite an apparently normal total PVR.

EXAMINER MARK SCHEME & MARKING GUIDELINES

TOPIC 9 – GAS EXCHANGE – Cambridge International AS Level Biology (9700)

Q	PART	EXAMINER MARKING PRINCIPLES & ACCEPTED ANSWERS	MARKS
1		Larynx → trachea → main / primary bronchus → lobar / secondary bronchus → bronchiole → terminal bronchiole → respiratory bronchiole → alveolar duct / sac / alveolus [2].	[2]
1		Incomplete dorsal gap allows the esophagus (located directly posterior) to expand during swallowing of a food bolus; while anterior/lateral cartilage prevents tracheal collapse during negative pressure generated on inspiration [2].	[2]
1		Pleural fluid acts as a lubricant reducing friction between visceral and parietal pleura during respiratory movements; surface tension adheres lungs to thoracic wall, ensuring lungs expand when thoracic volume increases [2].	[2]
2		Chondrocytes located within lacunae; embedded in a firm, hydrated extracellular matrix containing collagen fibrils and chondroitin sulfate proteoglycans [2].	[2]
2		Seromucous glands secrete watery fluid and mucin to trap airborne particles and humidify inspired air [1]; elastic fibres allow stretching during inspiration and recoil to maintain airway patency and elasticity [1].	[2]
2		Smooth muscle located in the posterior / dorsal gap of the cartilage C-ring [1]; contracts during the cough reflex to narrow the tracheal lumen, accelerating linear airflow velocity to expel mucus plugs [1].	[2]
3		Axoneme exhibits a '9+2' arrangement (9 peripheral microtubule doublets surrounding 2 central singlets) [1]; dynein arms hydrolyse ATP to cause sliding of adjacent doublets, producing a coordinated metachronal wave [1].	[2]
3		Aerobic respiration generates abundant ATP [1]; required by dynein ATPase arms for continuous ciliary motility and active transport at basal bodies [1].	[2]
3		Synthesised at rough ER, glycosylated and packaged in Golgi body, secreted via exocytosis of mucin granules [1]; traps dust, pollen, and bacteria, preventing microbial access to alveoli [1].	[2]
4		Bronchus contains irregular cartilage plates whereas bronchiole has no cartilage [1]; bronchus contains submucosal glands / goblet cells whereas terminal bronchiole lacks glands / has simple cuboidal epithelium [1].	[2]
4		Contraction of smooth muscle causes bronchoconstriction, reducing airflow to hypoventilated or damaged alveoli [1]; relaxation causes bronchodilation (e.g. stimulated by adrenaline), reducing airway resistance during exercise [1].	[2]
4		Without rigid cartilage, bronchiole lumen is held open solely by radial traction of surrounding elastic fibres; when elastase destroys elastic fibres, positive intrapleural pressure during forced expiration causes bronchioles to collapse, trapping air [2].	[2]
5		Squamous epithelial cell (Type I pneumocyte) cytoplasm / cell surface membrane [1]; fused extracellular basement membrane [1]; capillary endothelial cell cytoplasm / cell surface membrane [total < 0.5 µm path, 2].	[2]
5		Type I pneumocytes are extremely flattened squamous cells providing a minimal diffusion distance [1]; Type II pneumocytes are rounded / cuboidal secretory cells that synthesise and exocytose dipalmitoylphosphatidylcholine (surfactant) [1].	[2]
5		High surface tension of water lining alveoli remains unopposed; small alveoli collapse during expiration (atelectasis), requiring immense muscular effort to reinflate (infant respiratory distress syndrome) [2].	[2]
6		Fibres stretch as thoracic cavity expands, allowing alveoli to inflate without bursting [1]; stored potential energy causes passive elastic recoil during exhalation, forcing air out of alveoli [1].	[2]
6		Does not require muscular contraction or ATP hydrolysis; relies entirely on relaxation of diaphragm and external intercostals combined with elastic recoil of lung parenchyma [2].	[2]
6		Loss of alveolar recoil means lungs cannot fully empty; air remains trapped in alveoli, increasing residual volume and permanently hyperinflating the thoracic cavity (barrel chest deformity) [2].	[2]
7		Tidal Volume (VT) = 2.85 – 2.35 = 0.50 dm ³ [1]; Vital Capacity (VC) = 5.60 – 1.20 = 4.40 dm ³ (or VC = IRV + VT + ERV = 2.75 + 0.50 + 1.15 = 4.40 dm ³) [1].	[2]
7		Volume of air remaining in the lungs after a maximal forced expiration (~1.2 dm ³) [1]; cannot be measured because it cannot be exhaled into the spirometer chamber (requires helium dilution or plethysmography) [1].	[2]
7		PVR = Tidal Volume x Breathing Rate [1]; PVR = 0.50 dm ³ x 14 min ⁻¹ = 7.0 dm ³ min ⁻¹ (units required) [1].	[2]
8		Rate of diffusion is directly proportional to surface area (A) and concentration / partial pressure gradient (DeltaP), and inversely proportional to diffusion distance (x): Rate (A x DeltaP) / x [2].	[2]
8		Ventilation constantly replenishes O ₂ in alveoli, keeping alveolar pO ₂ high (~104 mmHg) [1]; continuous perfusion carries oxygenated blood away and delivers deoxygenated blood (pO ₂ ~40 mmHg), keeping capillary pO ₂ low [1].	[2]
8		Carbon dioxide has a diffusion constant / solubility in water and lipid membranes approximately 20 times higher than that of oxygen; higher solubility compensates for the lower concentration gradient [2].	[2]
9		Hypertrophy and hyperplasia of goblet cells and submucosal glands [1]; loss, paralysis, or destruction of cilia (ciliostasis) and squamous metaplasia [1].	[2]
9		Excessive, thick mucus cannot be removed by paralyzed cilia; accumulation of mucus triggers sensory cough receptors in bronchial walls to forcefully expel mucus plugs [2].	[2]

Q	PART	EXAMINER MARKING PRINCIPLES & ACCEPTED ANSWERS	MARKS
9		Stagnant, warm mucus pools in lower airways, providing a nutrient-rich culture medium; pathogens (e.g. <i>Streptococcus pneumoniae</i> , <i>Haemophilus influenzae</i>) are not cleared and colonise airway tissues [2].	[2]
10		Particulates are phagocytosed by alveolar macrophages, which release inflammatory cytokines (e.g. IL-8, TNF-alpha, leukotrienes) that attract circulating neutrophils across capillary walls [2].	[2]
10		alpha-antitrypsin is a serine protease inhibitor that inactivates elastase, protecting alveolar elastin [1]; oxidants in cigarette smoke oxidize active-site methionine residues of alpha-antitrypsin, abolishing its inhibitory capacity [1].	[2]
10		Alveolar breakdown drastically reduces total surface area for diffusion [1]; destruction of capillaries and loss of elastic recoil cause air trapping, leading to severe arterial hypoxemia and acidosis [1].	[2]
11		Stimulates nicotinic acetylcholine receptors and induces adrenal medulla to release adrenaline [1]; causes systemic arteriolar vasoconstriction, elevating peripheral resistance, heart rate, and arterial blood pressure [1].	[2]
11		Binds irreversibly to Fe ²⁺ in haem groups forming carboxyhaemoglobin (HbCO), reducing oxygen carriage; crosses placenta causing fetal hypoxia, leading to low birth weight or premature birth [2].	[2]
11		Sticky residue coats and paralyzes cilia, causing mucus stagnation [1]; contains carcinogens (e.g. benzo[a]pyrene) that form DNA adducts causing mutations in proto-oncogenes and tumor suppressor genes [1].	[2]
12		Carcinogen: an agent (chemical/physical) capable of inducing genetic mutations that cause cancer [1]; Metastasis: the spread of malignant tumor cells from primary bronchial site via blood or lymph to secondary organs [1].	[2]
12		Mutated proto-oncogenes (oncogenes, e.g. KRAS) code for hyperactive growth factor receptors stimulating continuous mitosis [1]; mutated tumor suppressor genes (e.g. TP53) fail to arrest cell cycle or trigger apoptosis in damaged cells [1].	[2]
12		Persistent hemoptysis (coughing blood) due to tumor erosion of bronchial blood vessels [1]; dyspnea / wheezing due to physical obstruction of bronchial lumen by growing tumor mass [1].	[2]
13		Diaphragm contracts and flattens downwards [1]; external intercostal muscles contract, pulling ribs upwards and outwards [1].	[2]
13		Thoracic volume increases; according to Boyle's law ($P \propto 1/V$), intrapulmonary pressure decreases below atmospheric pressure (-1 mmHg) [1]; air moves into the lungs down the pressure gradient until equilibrium is reached [1].	[2]
13		Forced expiration is active: internal intercostal muscles contract pulling ribs down and in [1]; abdominal wall muscles contract pushing abdominal viscera and diaphragm upwards, further reducing thoracic volume rapidly [1].	[2]
14		Elastic fibres [1]; allow expansion during inspiration to accommodate air volume and provide elastic recoil to drive passive exhalation and prevent alveolar over-distension [1].	[2]
14		Transitions from tall pseudostratified ciliated columnar (trachea) → simple columnar (bronchi) → simple ciliated cuboidal (bronchioles) → extremely thin simple squamous epithelium (alveoli) [2].	[2]
14		Goblet cell mucus would coat alveolar walls, vastly increasing diffusion distance and blocking gas diffusion [1]; rigid cartilage would prevent alveolar expansion and recoil and obstruct capillary perfusion [1].	[2]
15		Glass wool turns dark brown / yellow and becomes sticky [1]; due to the deposition and condensation of particulate tar [1].	[2]
15		Indicator turns yellow / orange / red (pH decreases / becomes acidic) [1]; due to dissolved oxides of nitrogen (NO _x), sulfur dioxide (SO ₂), or carbon dioxide forming acidic solutions (e.g. HNO ₃ , H ₂ SO ₃ , H ₂ CO ₃) [1].	[2]
15		Limewater turns milky / cloudy due to carbon dioxide reacting to precipitate calcium carbonate [1]; limitation: dry glass wool lacks living ciliated epithelium, moist mucus lining, blood supply, or macrophage immune responses [1].	[2]
16		Haemoglobin has an affinity for carbon monoxide approximately 200–250 times greater than its affinity for oxygen [1]; binding is essentially irreversible under standard physiological conditions, displacing oxygen [1].	[2]
16		Binding of CO locks haemoglobin in the high-affinity R-state, shifting the oxygen dissociation curve to the left [1]; prevents the unloading of oxygen to respiring tissues at physiological tissue pO ₂ , causing cellular hypoxia [1].	[2]
16		Chronic tissue hypoxia stimulates renal cells to secrete erythropoietin (EPO) [1]; EPO stimulates bone marrow erythropoiesis to produce more erythrocytes (secondary polycythemia), increasing blood viscosity [1].	[2]
17		Chronic bronchitis primarily affects conducting airways (bronchi and larger bronchioles) [1]; emphysema primarily affects respiratory zones (alveolar walls, septa, and respiratory bronchioles) [1].	[2]
17		Emphysema patients maintain near-normal arterial pO ₂ by hyperventilating (pink appearance) [1]; chronic bronchitis patients suffer severe ventilation-perfusion mismatch, leading to chronic cyanosis / low pO ₂ (blue) and fluid retention / oedema (bloating) [1].	[2]
17		Chronic alveolar hypoxia causes widespread pulmonary arteriolar vasoconstriction and loss of capillary beds, vastly increasing pulmonary vascular resistance [1]; right ventricle must generate higher pressure to pump blood into pulmonary circuit, leading to RV hypertrophy and right-sided failure [1].	[2]
18		$SA:V = 6I^2 / I^3 = 6 / I$ [1]; as side length I increases, volume increases proportionally to I^3 while surface area increases only to I^2 , causing SA:V ratio to decrease [1].	[2]

Q	PART	EXAMINER MARKING PRINCIPLES & ACCEPTED ANSWERS	MARKS
18		Diffusion rate / speed of H ⁺ ion movement depends on temperature and concentration gradient, not cube size [1]; but in the largest cube, diffusion distance to center is too great, so central core remains unpenetrated [1].	[2]
18		Mammals have a very small SA:V ratio and large internal diffusion distances (> several cm) [1]; diffusion across body surface is far too slow to supply high metabolic oxygen demands; lungs provide a massive internal surface area (~100 m ²) with minimal diffusion distance (< 0.5 μm) [1].	[2]
19		Correlation merely indicates two variables change together; could be coincidental or driven by confounding third variables (e.g. industrial air pollution, genetic predisposition) [2].	[2]
19		Identification of specific chemical carcinogens (benzo[a]pyrene) in tar [1]; demonstration that tar induces tumors in animal skin painting experiments, and mutates codons 175, 248, 273 of human TP53 gene in vitro [1].	[2]
19		Carcinogenesis is a multi-step progressive process requiring multiple cumulative mutations (5-6 mutations in oncogenes/tumor suppressors) over decades before a cell escapes growth control and forms a detectable tumor [2].	[2]
20		Hydrophilic phosphate / choline head groups face downwards into the aqueous alveolar fluid lining [1]; hydrophobic fatty acid tails project upwards into the alveolar air space, disrupting water hydrogen bonding [1].	[2]
20		If surface tension (T) were uniform, inward collapsing pressure (P) would be inversely proportional to radius (r); smaller alveoli would have higher internal pressure, emptying their air into larger alveoli and collapsing (atelectasis) [2].	[2]
20		Endotracheal instillation of exogenous / synthetic animal surfactant [1]; continuous positive airway pressure (CPAP) ventilation or maternal corticosteroid injection prior to preterm delivery [1].	[2]
21		Trachea: C-shaped (incomplete) rings [1]; Bronchi: irregular curved plates / islands of cartilage [1].	[2]
21		Bronchioles need to constrict and dilate their lumen dynamically to regulate ventilation [1]; rigid cartilage would prevent diameter changes, and bronchioles are kept patent by radial tension of surrounding lung parenchyma [1].	[2]
22		Cilium: core of microtubules arranged in a '9+2' axoneme arising from a basal body [1]; Microvillus: core of actin microfilaments bundled together with no basal body [1].	[2]
22		Cilia are motile organelles that beat to sweep mucus and trapped pathogens upwards out of the airway [1]; microvilli are non-motile folds that increase surface area for absorption of digested nutrients [1].	[2]
23		In response to histamine / allergens, smooth muscle contracts spasmodically (bronchospasm) [1]; narrowing the lumen and massively increasing resistance to airflow [1].	[2]
23		Elastic fibres cannot actively contract; they are stretched by external forces during inspiration and recoil passively due to elastin protein elasticity [1]; smooth muscle contains actin/myosin and contracts actively via ATP hydrolysis [1].	[2]
24		Mucins are large glycoproteins rich in oligosaccharide chains and negatively charged sialic acid / sulfate [1]; upon exocytosis, they bind large amounts of water to form a viscous gel [1].	[2]
24		Mucus accumulates in bronchioles forming plugs, blocking ventilation to downstream alveoli [1]; creates a physiological shunt where blood flows through unventilated capillaries without taking up oxygen [1].	[2]
25		Millions of spherical alveoli (approx. 500-700 million in adult lungs) [1]; highly folded, irregular inner alveolar surface and dense surrounding capillary meshwork [1].	[2]
25		Capillaries are in direct contact with alveolar epithelium, with basement membranes fused together [1]; capillary diameter is narrow (~7 μm), forcing erythrocytes to travel in single file and press against endothelium [1].	[2]
26		Inspired: ~20.9% O ₂ , ~0.04% CO ₂ [1]; Expired: ~16.0% O ₂ , ~4.0% CO ₂ [1].	[2]
26		Only a fraction of alveolar air is replaced with each tidal breath (~0.5 dm ³ out of ~3.0 dm ³ FRC) [1]; inspired air mixes with anatomical dead space air (trachea, bronchi) which does not undergo gas exchange [1].	[2]
27		Stimulates sympathetic nervous system and triggers adrenaline release from adrenal glands [1]; adrenaline increases sinoatrial node firing rate and constricts peripheral arterioles [1].	[2]
27		Stimulates platelet activation and increases platelet stickiness, promoting thrombus formation [1]; hypertension damages endothelial lining, initiating LDL deposition and atheromatous plaque development [1].	[2]
28		Carboxyhaemoglobin (HbCO) [1]; bond energy is much higher / dissociation rate constant is extremely low compared to oxyhaemoglobin [1].	[2]
28		Effective oxygen-carrying capacity of blood is reduced by 20% [1]; remaining Hb binds O ₂ too tightly (left shift), starving active muscle mitochondria of oxygen for aerobic ATP synthesis [1].	[2]
29		Ciliostasis is the cessation / paralysis of ciliary beating [1]; toxic chemicals in tar inhibit dynein ATPase arms and damage axonemal microtubules [1].	[2]
29		The mucociliary escalator fails; contaminated mucus remains trapped in the airways, pooling in bronchi and providing a focus for bacterial infection [2].	[2]
30		Hypertrophy: increase in individual cell size and secretory volume [1]; Hyperplasia: increase in cell number due to mitotic division of basal cells [1].	[2]
30		Accumulation of thick mucus plugs inside lumen [1]; combined with mucosal edema / swelling from chronic inflammation and smooth muscle hypertrophy [1].	[2]

Q	PART	EXAMINER MARKING PRINCIPLES & ACCEPTED ANSWERS	MARKS
31		Alveolar septa are broken down / degraded, resulting in fewer, much larger, irregularly shaped air sacs (bullae) [2].	[2]
31		Larger air spaces have a drastically lower surface area-to-volume ratio (total SA falls from ~100 m ² to < 30 m ²) [1]; according to Fick's law, reduced surface area proportionally decreases diffusion rate [1].	[2]
32		Neutrophil (or alveolar macrophage) [1]; releases elastase to break down extracellular matrix and bacterial proteins during phagocytosis / migration [1].	[2]
32		Lack the circulating inhibitor that neutralises elastase [1]; basal physiological elastase released by routine inflammatory turnover digests elastin unchecked over time [1].	[2]
33		Metabolised into an epoxide intermediate that covalently binds to guanine bases, forming DNA adducts [1]; causes mispairing during DNA replication leading to transversion mutations [1].	[2]
33		p53 protein normally detects DNA damage and induces G1 cell cycle arrest or apoptosis [1]; mutated p53 allows damaged cells to divide uncontrollably by mitosis, forming a tumor [1].	[2]
34		Diaphragm flattens downwards from a dome shape [1]; increases vertical / longitudinal dimension and volume of thoracic cavity [1].	[2]
34		Increased volume causes intrapulmonary pressure to drop below atmospheric pressure [1]; creating a pressure gradient down which air flows into lungs by bulk flow [1].	[2]
35		None / no muscles contract actively; it is purely passive, driven by relaxation of external intercostals/diaphragm and elastic recoil of lungs [2].	[2]
35		Internal intercostal muscles contract, pulling ribs forcibly down and in [1]; abdominal wall muscles (rectus abdominis, obliques) contract, forcing diaphragm rapidly upwards [1].	[2]
36		Polar water molecules form cohesive hydrogen bonds with each other, pulling surface molecules inward towards the liquid phase [2].	[2]
36		Surfactant phospholipids intersperse between water molecules, reducing intermolecular attractive forces [1]; prevents alveolar collapse (atelectasis) and reduces muscular effort required to inflate lungs [1].	[2]
37		The volume of air inhaled or exhaled during a single normal, quiet breath at rest [1]; approximately 0.5 dm ³ (500 cm ³) [1].	[2]
37		The maximum volume of air that can be exhaled following a maximal forced inspiration [1]; VC = Inspiratory Reserve Volume (IRV) + Tidal Volume (VT) + Expiratory Reserve Volume (ERV) [1].	[2]
38		Transit time ~0.75 s at rest [1]; steep initial partial pressure gradient (DeltaP = 64 mmHg) and minimal diffusion distance (< 0.5 µm) allow complete saturation within first third of capillary length [1].	[2]
38		Slows linear velocity of RBCs, maximising contact time [1]; presses erythrocyte membrane against endothelial cell, minimising diffusion pathway through plasma layer [1].	[2]
39		Mainstream smoke is exhaled by the smoker; sidestream smoke rises directly from burning cigarette tip [1]; sidestream burns at lower temperature, containing higher concentrations of toxic gases (CO, ammonia, carcinogens) [1].	[2]
39		Increased frequency and severity of asthma attacks [1]; higher incidence of lower respiratory tract infections (bronchitis, pneumonia) and otitis media (middle ear infection) [1].	[2]
40		Originate from blood monocytes that migrate into lung parenchyma and patrol the luminal surface of alveolar septa [2].	[2]
40		Engulf particulates by phagocytosis and store them in lysosomes [1]; when overwhelmed by heavy smoke, they undergo lysis, releasing lysosomal proteases that damage alveolar walls and provoke chronic inflammation [1].	[2]
41		The physical process by which oxygen diffuses from alveolar air into blood and carbon dioxide diffuses from blood into alveolar air across a specialised respiratory surface [2].	[2]
42		Ventilation is the mechanical bulk flow of air into and out of lungs driven by pressure changes [1]; cellular respiration is the intracellular biochemical breakdown of glucose to yield ATP [1].	[2]
43		Keeps the tracheal airway patent / open, preventing collapse during inspiration when intrathoracic pressure is negative [2].	[2]
44		Undergo passive elastic recoil following stretch, forcing air out of alveoli without muscular effort and returning alveoli to resting volume [2].	[2]
45		The volume of air inspired or expired in a single normal resting breath [1]; ~0.5 dm ³ (500 cm ³) [1].	[2]
46		The maximum volume of air that can be exhaled from the lungs following a maximal forced inhalation [2].	[2]
47		Persistent productive cough with excessive sputum production for at least 3 consecutive months in 2 successive years [2].	[2]
48		Permanent, irreversible destruction and enlargement of alveolar walls/septa with loss of elastic fibres and reduced surface area [2].	[2]
49		A stable complex of carbon monoxide bound to the iron(II) haem groups of haemoglobin [1]; permanently prevents oxygen binding and transport [1].	[2]

Q	PART	EXAMINER MARKING PRINCIPLES & ACCEPTED ANSWERS	MARKS
50		An environmental agent (chemical, physical, biological) that causes genetic mutations leading to cancer [1]; benzo[a]pyrene / polycyclic aromatic hydrocarbons / nitrosamines [1].	[2]